TEXORA · SESSIONAL NOTES · STUDY SHEET

Pattern Grading for Trouser

Grading the trouser block S–XL by cardinal-point increments derived from the chart's grade rules, and verifying every graded size by measurement against the same chart.

TE-3108 · APPAREL MANUFACTURING TECHNOLOGY-II SESSIONAL · STUDY 6 · LEVEL 3 TERM 1

1. Aim and place of the study

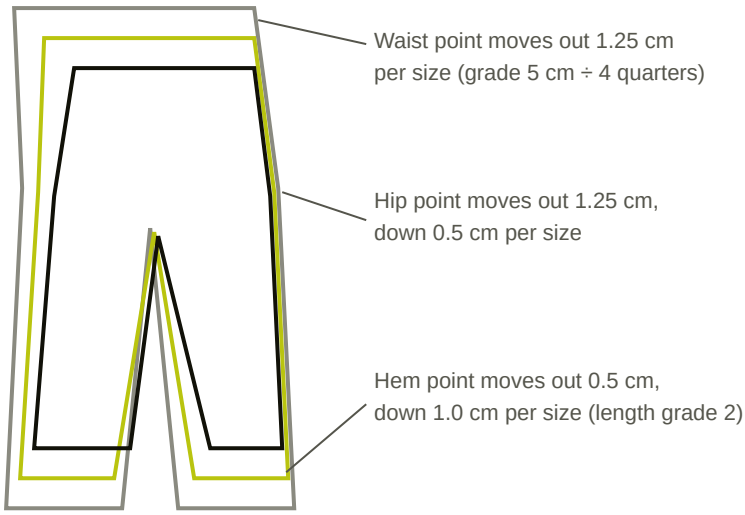
Aim: to grade the basic trouser front (and by the same rules the back) from master size M to sizes S, L and XL by the increment (grade-rule) method, and to verify the graded nest against the TE-3108 chart at every principal point.

2. Materials, machine settings and specimens

Materials: Study 4 front piece (size M), grading ruler/square, pattern paper, coloured pencils per size. **Grade rules from the chart:** waist 5 cm and hip 5 cm per size → per quarter panel: 5 ÷ 4 = **1.25 cm** in width; length (outseam) grade 2 cm between S–L (1 cm at XL step) applied at the hem; thigh, knee and hem widths grade 2 cm → 0.5 cm per side about the crease; rise grades 1 cm.

Trouser measurement chart used throughout TE-3108 (men's basic trouser, garment values in cm): waist S 77 · **M 82** · L 87 · XL 92 (grade 5); hip 97 · **102** · 107 · 112 (grade 5); outseam (incl. waistband) 102 · **104** · 106 · 107; inseam 76 · **78** · 80 · 81; thigh 60 · **62** · 64 · 66 (grade 2); knee 44 · **46** · 48 · 50 (grade 2); hem (bottom) width 38 · **40** · 42 · 44 (grade 2); front rise 25 · **26** · 27 · 28. Size M is the working size unless the study states otherwise. Seam allowances, as in TE-2207/TE-2208: **1 cm** general, **1.5 cm** at side seams, waistband and other adjustment seams, **2.5 cm** at hems. The shirt values of TE-2207/2208 (size M chest 107, length 73, shoulder 46, sleeve 61.5, collar 40; grades 5/2/2/1.5/2 cm) are used unchanged wherever a shirt is the specimen.

3. Cardinal points and their movements (per size step, from M)



Nested nest of S (outer grey) · M (green) · L (black): every cardinal point moves by its own (x, y) grade rule.

Figure 1 — Grading a trouser front: the master size M and its neighbours, with the movement of three cardinal points.

POINT	X (WIDTH) MOVEMENT	Y (LENGTH) MOVEMENT
Waist at side seam	Out 1.25 (quarter of waist grade 5)	0
Hip at side seam	Out 1.25 (quarter of hip grade 5)	Down 0.5
Centre front at waist	Out 0.3 (share of waist grade)	Up 0.25 of rise grade disposition
Crutch point	Out 0.6	Down 0.5 (rise grade 1 disposed)
Knee, each side of crease	Out 0.5	Down 1.0
Hem, each side of crease	Out 0.5	Down 2.0 (full length grade at the hem)

4. Procedure and verification

1. Fix the front piece; mark the cardinal points and the crease line as the grading origin reference (the crease does not move in width grading).
2. Step each point by (x, y) once for L and again for XL; reverse the signs for S. Join the stepped points with curves parallel in character to the master's.
3. Trace each size as a separate piece with its own information block (size S / L / XL).
4. **Verify the nest against the chart:** measure the graded L front: quarter waist must read 87 ÷ 4 = 21.75 cm (master 20.5 + 1.25 ✓); quarter hip 107 ÷ 4 = 26.75 (25.5 + 1.25 ✓); outseam 106 (104 +

2 ✓); front hem width 21 cm (20 + 0.5 per side × 2; the full leg hem therefore reads 42). Any point failing its chart value is re-stepped, not 'adjusted by eye'.

Precautions

- Grade from the master, never size-from-size: stepping S from M and L from S compounds errors.
- Keep each size's pencil colour consistent through points, curves and tracing.
- The dart on the back grades in position with the waist points; its intake stays 2 cm across this size range unless the chart says otherwise.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Calculate the per-quarter width increment for a girth grading 5 cm, and the graded quarter-hip of size XL (hip 112).
2. Why is the full length grade applied at the hem rather than shared between knee and hem?
3. A graded size S front measures quarter waist 19.6 cm; the chart expects 77 ÷ 4 = 19.25. What happened?

Answer key

1. 5 ÷ 4 = 1.25 cm per quarter per step. XL is two steps above M: 25.5 + 2 × 1.25 = 28.0 cm (= 112 ÷ 4 ✓).
2. Body length growth is taken below the knee in this system so the knee position and rise proportions of the master stay constant; sharing it would move the knee line and change the garment's balance.
3. The S was stepped with the wrong sign or magnitude at the waist points — most likely stepped from the L side or stepped 1.25 twice in the same direction at one point. Re-step from the master M with reversed signs and re-measure.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib